

Sarat Thokala

Senior Architect • **Agentic AI • AI Engineering • LLMs • Cloud (AWS, Azure, GCP)**
• Hyderabad, India • 20 Years  <https://www.linkedin.com/in/sarat-thokala/>

PROFESSIONAL SUMMARY

Senior Architect with **20 years of experience** in product engineering, enterprise architecture, and cloud-native application development. Expertise in designing scalable AI-powered enterprise solutions using **Python, Node.js, FastAPI, LLMs, Agentic AI, AWS Bedrock, and Azure AI Foundry**. Strong experience in architecting **microservices, event-driven, and multi-tenant SaaS** applications across AWS and Azure. Backed by a solid foundation in enterprise analytics, Java/J2EE distributed systems, and technical leadership, delivering secure, scalable, production-ready solutions.

CORE SKILLS

AI & Generative AI	LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, AI agents, Prompt Engineering, OpenAI, CrewAI, AWS Bedrock, Azure AI, Langgraph, Langchain
Python & Backend	Python, FastAPI, Async Programming, REST APIs
Architecture	Enterprise Architecture, Microservices, Domain-Driven Design (DDD), Event-Driven Architecture, SaaS Multi-Tenant Architecture
Cloud Platforms	AWS (Lambda, EC2,S3, IAM, ECS, RDS, Secrets Manager, CloudWatch) Azure (Key Vault, Blob Storage, REST APIs, VMs)
Cloud Analytics	AWS Glue, Amazon Athena, Quicksight, ETL Pipelines, Amazon Q
Full-Stack tech	Node.js, TypeScript, Express.js, Sequelize, Java/J2EE, Spring MVC
Databases	PostgreSQL, MongoDB, MySQL, Oracle, PL/SQL
Message/ Streaming Dev Ops/ Tools	Apache Kafka, AWS MSK, NATS, Docker, Kubernetes, CI/CD, Unix Shell Scripting

PROFESSIONAL EXPERIENCE

Senior Architect – AI Engineering / Product Engineering

Trianz • Hyderabad, India • **July 2023 – Present**

- **Architected** enterprise Discovery, and Cloud Analytics platforms by integrating AI-powered automation, cloud-native microservices, and distributed event-driven architectures to improve infrastructure discovery, operational intelligence, and cloud migration insights.
- Designed and implemented **Agentic AI** solutions using **AWS Bedrock** and **Azure AI Foundry** for intelligent workflow automation.
- Designed and developed AI agents and multi-agent workflows using OpenAI, CrewAI, LangGraph, and LangChain.
- Integrated LLMs for automated log analysis, root-cause identification, remediation recommendations, and intelligent operational insights.
- Architected and developed scalable Python/FastAPI microservices using asynchronous programming and event-driven design principles.
- Designed high-throughput messaging solutions using Apache Kafka, AWS MSK, and NATS for real-time distributed processing.
- Led solution architecture, technical roadmap, engineering best practices, code reviews, and cross-functional technical leadership.

- Drove end-to-end product delivery from architecture and design through implementation, deployment, and production support.

Environment: Python, FastAPI, OpenAI, CrewAI, LangGraph, LangChain, LLMs, Agentic AI, RAG, AWS Bedrock, Azure AI Foundry, Apache Kafka, AWS MSK, NATS, PostgreSQL, Docker, Kubernetes, AWS SDK, Azure REST APIs

Architect – Enterprise Platform (CMDB)

Trianz • Hyderabad, India • Jan 2021 – June 2023

- Architected and delivered an enterprise-scale, cloud-native CMDB platform from the ground up, integrating AWS, Azure, GCP, and VMware using cloud SDKs and REST APIs.
- Designed a scalable multi-tenant SaaS architecture supporting both cloud-hosted and customer-managed private deployments.
- Engineered a high-performance parallel data ingestion framework with automated validation, transformation, and relationship mapping for large-scale cloud resource discovery.
- Designed a metadata-driven schema management framework enabling dynamic model creation, runtime schema evolution, and automated database migrations.
- Developed an intelligent relationship discovery and identity reconciliation engine to ensure accurate configuration item mapping and data integrity.
- Led solution architecture, technical design, engineering standards, and end-to-end product delivery across cross-functional teams.

Environment: Python, Node.js, TypeScript, Express.js, Sequelize ORM, PostgreSQL, AWS SDK, Azure REST APIs, Apache Kafka (AWS MSK), NATS, Docker, Kubernetes

Senior Technical Lead – Athena Federated Query (AFQ)

Trianz • Hyderabad, India • July 2020 – Dec 2020

- Built AFQ connectors with Java enabling SQL analytics across Hive and Impala without data migration
- Deployed connectors to AWS Marketplace as Lambda functions with automated scaling and S3 spillover
- Conducted performance testing with JProfiler for optimised query execution and resource utilisation

Environment: Java, AWS Athena, AWS Lambda, S3, Hadoop, Hive, Impala, JProfiler

Big Data Solution Architect

Synchrony Financial (via Capgemini) • Hyderabad, India • Aug 2019 – July 2020

- Designed Spark-Scala ingestion framework loading customer data from SQL Server, HDFS, and file systems
- Applied Protegrity encryption rules on sensitive PII data across Raw and Curation layers
- Developed curation pipelines with transformation rules; scheduled batch processing via Autosys
- Created Sqoop jobs for RDBMS-to-Hive data imports and guided team on technical solutions

Environment: Hadoop 2.x, Spark 2.0, Scala, Kafka, Hive, Sqoop, Autosys, UNIX Shell Scripts, PL/SQL, Jenkins, GIT

Big Data Technical Lead

HCSC Inc. (via Infosys) • Hyderabad, India • May 2018 – July 2019

- Built multi-layer healthcare data platform (Raw, Merge, Gold) for claims, member, and health care data
- Implemented Kafka consumer/producer for near-real-time processing alongside scheduled batch loads
- Applied ORC compression, Spark transformations, and Hive merge scripts for data quality and performance

Environment: Hadoop 2.x, Spark 2.0, Scala, Kafka, Hive, Sqoop, Zena Scheduler, UNIX Shell Scripts, Talend, Jenkins

Java Architect

Hibbett Sports Inc. (via Infosys) • Birmingham, USA • Feb 2016 – Apr 2018

- Developed Spark SQL and Spark Core batch jobs for e-commerce, POS, and CRM transaction analytics
- Provisioned multi-tenant Hadoop clusters on AWS (EC2, S3) with Oozie automation and Zookeeper coordination
- Loaded subscriber data from MSSQL and DB2 via Sqoop; exported analysed data to RDBMS for BI reports
- Acted as onsite coordinator — managed offshore team and provided client status updates

Environment: Hadoop 2.x, Spark 2.0, Scala, Hive, Kafka, Sqoop, Oozie, MapReduce, HBase, AWS (EC2, S3, RDS), Jenkins

Java Technology Lead

Gap Inc. (via Infosys) • San Francisco, USA & Pune, India • **July 2010 – Jan 2016**

- Maintained and enhanced J2EE enterprise POS applications: ORPOS, SIA, DPP, FMR, Shipment, ORCO
- Led WebSphere-to-JBoss migration for all retail web applications with zero downtime
- Developed MapReduce programs and Hive/HBase pipelines for retail sales analytics and marketing reports
- Enhanced FMR web application for Athleta brand; implemented IVR Lotto Calendar in POS for Puerto Rico
- Led 8-member team; prepared technical design documents and mentored new joiners

Environment: Java, J2EE, EJB, JMS, JSP, Servlets, Spring MVC, Hadoop, MapReduce, Hive, HBase, Pig, Sqoop, JIRA, Oracle, PL/SQL

Java Developer

Tesco (via Infosys) • Bangalore, India • **Jan 2010 – Jun 2010**

- Implemented change requests for ORPM (Oracle Retail Price Management) — New Buying System for UK and ROI
- Worked on promotion management, complex deal configuration, and promotional forecasting features
- Implemented unit test cases; worked extensively with EJB, Hibernate, DAO, and Swing UI

Environment: Java, EJB, Hibernate, PL/SQL, oc4j, Oracle ORPM, ClearCase

Java Developer

Discover Financial Services (via Capgemini) • Pune, India • **Jan 2009 – Sep 2009**

- Migrated credit card call centre application (NEIL) from ATG server to JBoss, replacing ATG-specific components with JEE-compliant equivalents
- Upgraded EJB 1.0 components to EJB 3.0; implemented Quartz scheduling mechanisms
- Created UML sequence and use case diagrams using Rational Rose; coordinated with mainframe teams

Environment: Java, JEE, JBoss, EJB 3.0, Quartz, Struts, JSP, Servlets, JAXB, JPA, WebSphere, Rational Rose

Java Developer

HSBC (via Capgemini) • Hyderabad, India • **Apr 2006 – Dec 2008**

- Developed and customised eChamps — a credit card call centre application serving call centre representatives globally
- Implemented features for card blocking, transaction history, funds transfer, loyalty points, and chargeback requests
- Built internationalisation capabilities enabling application deployment across multiple geographic regions
- Developed server-side components using Servlets, EJB, and JMS; built UI with JSP, JSTL, and JavaScript
- Prepared Maven build scripts and deployed to WebSphere application server

Environment: Java, J2EE, WSAD, JSP, Servlets, EJB, JMS, JSTL, JavaScript, HTML, Maven, WebSphere, MKS, DS2

CERTIFICATIONS & RECOGNITION

AWS Certified AI Practitioner

AWS Certified Data Analytics – Specialty

AWS Certified Developer – Associate

Sun Certified Programmer for Java 2 Platform

Awards

- **Trianz Global Leadership Award** — Trend Setter, Growth & Profitability (2021) — Awarded by the President of Trianz for outstanding contributions to client satisfaction and delivery excellence
- **Individual Excellence Spot Award (IESA)** — Trianz, Dec 2021 — Recognised by Chairman & CEO for ownership of CMDb and AFQ modules and direct client troubleshooting

EMPLOYMENT HISTORY

Trianz Limited • July 2020 – Present

Infosys Limited • Sep 2009 – Aug 2019

Capgemini Technology Services • Apr 2005 – Sep 2009, Aug 2019 – July 2020

EDUCATION

B. Tech, Computer Science & Information Technology

